

BISE Results Database (3NF Normalized Schema)

1. Entity: Exams

Primary Key (PK)

- exam_id

Table Columns

1. **exam_id**
 - Unique identifier for each exam
2. **exam_name**
 - Name of the examination (e.g., Matric, Intermediate)
3. **exam_year**
 - Year in which the exam was conducted
4. **exam_type**
 - Type of exam (Annual / Supplementary)

Purpose

Stores unique exam-related information and avoids repetition of exam data.

2. Entity: Subject_Master

Primary Key (PK)

- subject_id

Table Columns

1. **subject_id**

- Unique identifier for each subject
- 2. **subject_name**
 - Name of the subject (e.g., Mathematics, Physics)
- 3. **subject_type**
 - Category of subject (Compulsory / Elective)
- 4. **max_marks**
 - Maximum marks allocated to the subject
- 5. **has_practical**
 - Indicates whether the subject includes a practical exam

Purpose

Maintains standardized subject information and supports normalization.

3. Entity: Student_ Types

Primary Key (PK)

- type_id

Table Columns

1. **type_id**
 - Unique identifier for student type
2. **type_name**
 - Type of student (Regular / Private)
3. **description**
 - Additional details about the student type

Purpose

Separates student categories to reduce redundancy and improve clarity.

4. Entity: Status_ Types

Primary Key (PK)

- status_id

Table Columns

1. **status_id**
 - Unique identifier for each status
2. **status_name**
 - Result status (Pass, Fail, Absent)
3. **description**
 - Brief explanation of the status

Purpose

Stores result statuses in a normalized form for consistency.

5. Entity: Students_ Normalized

Primary Key (PK)

- roll_number

Table Columns

1. **roll_number**
 - Unique roll number assigned to each student
2. **student_name**
 - Full name of the student
3. **father_name**
 - Father's name of the student

4. **institute_id (FK)**

- References the institute where the student is enrolled

5. **type_id (FK)**

- References Student_ Types table

Purpose

Stores student personal and academic identification data in a clean, normalized structure.

6. Entity: Results

Primary Key (PK)

- result_id

Foreign Keys (FK)

- roll_number
- exam_id
- subject_id
- status_id

Table Columns

1. **result_id**

- Unique identifier for each result record

2. **roll_number (FK)**

- References Students_ Normalized

3. **Exam_id (FK)**

- References Exams

4. **subject_id (FK)**

- References Subject_ Master

5. **marks_obtained**

- Marks obtained by the student

6. **status_id (FK)**

- References Status_ Types

Purpose

Stores students' exam results while maintaining full referential integrity.

Overall Benefits of This Design

- Fully **3NF compliant**
 - Eliminates data redundancy
 - Improves data integrity and consistency
 - Easy to maintain and scale
 - Suitable for academic projects and real-world databases
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ERD DIAGRAM:

